Modern Lectures on QFT

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Abstract

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1 Scale-Invariant Physics

These notes summarize the Wilsonian renormalization-group viewpoint and its application to the Ising universality class. The prose has been lightly reconstructed; the equations have been normalized and checked for consistency.

1.1 Wilsonian RG in QFT

Consider a Euclidean scalar theory with cutoff Λ:

$$\mathcal{L} = \frac{1}{2}\big(\partial_\mu \varphi\big)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{\lambda}{4!}\varphi^4, \tag{1}$$

$$Z = \int [\mathcal{D}\varphi]_\Lambda \exp\!\Big(-\int d^d x \mathcal{L}[\varphi]\Big), \qquad [\mathcal{D}\varphi]_\Lambda = \prod_{|k|<\Lambda} d\varphi(k), \tag{2}$$

with Fourier convention

$$\varphi(x) = \int \frac{d^d k}{(2\pi)^d} e^{ik\cdot x} \varphi(k). \tag{3}$$

One may worry about changing into the fourier space may cause a change of measure, indeed, yet long as we focus on physical observables, the change of measure does not affect the final results.

> **Remark**
>
> Here in Wilsonian's picture, all QFT are defined up to a cut off in the momentum space or equivalently a lattice spacing in the position space.

1.1.1 Integrating out UV Modes

Now we want to consider how a theory changes as we lower the energy cut off. Now lower the cutoff from Λ to $\rho\Lambda$, where $0 < \rho < 1$, and split

$$\varphi(k) = \varphi_{\mathrm{IR}}(k) + \hat{\varphi}(k), \tag{4}$$

where φ_{IR} has support for $|k| < \rho\Lambda$ and $\hat{\varphi}$ has support for $\rho\Lambda \le |k| < \Lambda$. Then we assume that m, λ are small and we integrate out the high-momentum modes $\hat{\varphi}$:

$$Z = \int [\mathcal{D}\varphi]_{\rho\Lambda} \int [\mathcal{D}\hat{\varphi}] \exp\!\Big(-\int d^d x \mathcal{L}[\varphi + \hat{\varphi}]\Big) = \int [\mathcal{D}\varphi]_{\rho\Lambda} \exp\!\Big(-\int d^d x \mathcal{L}_{\mathrm{eff}}[\varphi]\Big). \tag{5}$$

The effective Lagrangian may in principal contains every local operator allowed by the symmetries:

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}(1+\Delta Z)(\partial\varphi)^2 + \frac{1}{2}(m^2+\Delta m^2)\varphi^2 + \frac{1}{4!}(\lambda+\Delta\lambda)\varphi^4 + \Delta C(\partial\varphi)^4 + \Delta D\varphi^6 + ... \quad (6)$$

All induced coefficients depend on the blocking parameter ρ. Now we see that the original theory if we integrate out the high momentum modes first we generates an equivalent theory with a lower cutoff and new lagrangian. Notice that the new lagrangian is not a completely new theory but rather a effective theory of the original theory if we only consider the low energy physics.

1.1.2 Flow on Coupling Space

Now we want to interpret the new effective theory as a theory with a change of the coupling from the original theory (while preserving the cut off and the Kinetic term). To do this we need to restore the original cutoff by defining

$$x' = \rho x, \quad k' = \frac{k}{\rho}, \quad \partial_\mu = \rho\partial'_\mu. \quad (7)$$

We also want the Kinetic term to be canonically normalized thus we can fix a normalization of the field and only have the coupling to flow.

$$\varphi' = \sqrt{\rho^{2-d}(1+\Delta Z)}\varphi. \quad (8)$$

The action therefore becomes

$$\int d^dx' \left[\frac{1}{2}(\partial'\varphi')^2 + \frac{1}{2}m'^2\varphi'^2 + \frac{\lambda'}{4!}\varphi'^4 + C'(\partial'\varphi')^4 + ...\right], \quad (9)$$

where

$$\begin{aligned} m'^2 &= (m^2+\Delta m^2)(1+\Delta Z)^{-1}\rho^{-2}, \\ \lambda' &= (\lambda+\Delta\lambda)(1+\Delta Z)^{-2}\rho^{d-4}, \\ C' &= (C+\Delta C)(1+\Delta Z)^{-2}\rho^d. \end{aligned} \quad (10)$$

> **Remark**
>
> Notice that until now we have not made any approximation. But obviously the coefficients for the new effective lagrangian are quite impossible to compute.

However, now we focus on leading order in m^2, λ, C and we have:

$$m'^2 = \rho^{-2}m^2, \quad \lambda' = \rho^{d-4}\lambda, \quad C' = \rho^d C. \quad (11)$$

We may see that different couplings behave differently under the RG flow. The mass term m^2 grows as we lower the cutoff, the λ term grows for $d < 4$ and shrinks for $d > 4$, and the C term shrinks for all $d > 0$. Thus we call:

- Relevant operators: m^2 and λ for $d < 4$.
- Irrelevant operators: C and λ for $d > 4$.
- Marginal operators: λ for $d = 4$. In fact this need detailed analysis, in dim reg we in fact work in $d = 4 - \varepsilon$ and we find that λ is marginally relevant for $\varepsilon > 0$.

1.1.3 RG Flow and Fixed Points

The flow of the couplings can be described by a function called the beta function:

$$\mu\frac{dg}{d\mu} = \beta(g). \quad (12)$$

with μ the energy scale and g the coupling. We may find out that there are some fixed point of the RG flow, which have some interesting behaviors.

YL: [Diagram in hand written note!]

1.2 RG in StatMech

For RG in StatMech, it is much more calculable compared to QFT (where in fact we seldome directly calculate the RG flow of Wilsonian coupling.) Now we take Ising model as a simplest model for an illustration. For lattice with spins $s_i \in \{+1, -1\}$ on each site:

$$H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i, \qquad P[\{s\}] = \frac{1}{Z}e^{-\beta H[\{s\}]}, \qquad Z = \sum_{[\{s\}]} e^{-\beta H[\{s\}]}, \tag{13}$$

where $\beta = \frac{1}{T}$. At $h = 0$, the low-temperature phase has nonzero spontaneous magnetization, while the high-temperature phase is disordered. Near the critical temperature,

$$M \sim (T_c - T)^\beta \tag{14}$$

on the ordered side. Here the exponent β should not be confused with inverse temperature. (This is experimentally true and we will see how to understand this.)

1.2.1 Block Spin Transformation

In RG of QFT we integrate out the high momentum modes and see how the theory changes. Now in SM, it is much convenient to stay in position space and we can make the lattice spacing larger and see how the theory changes. To do this we integrate out the blocks of spins to make it a coarse-grained lattice. This is realized by **Block Spin Transformation**:

$$T\big(s'; s_1, ..., s_q\big) = \begin{cases} 1 \text{ if } s'\sum_{i=1}^q s_i > 0 \\ 0 \text{ otherwise} \end{cases}. \tag{15}$$

This means that if most of the spins in a block are up, then the coarse-grained spin is up, and vice versa or we throw away the contribution.

YL: [Diagram for illustration!]

The coarse-grained Hamiltonian is defined by

$$e^{-\beta H'[\{s'\}]} = \sum_{[\{s\}]}\prod_{\text{blocks}} T\big(s'; s_1, ..., s_q\big)e^{-\beta H[\{s\}]}. \tag{16}$$

and the new partition function is

$$\begin{aligned} Z' &= \sum_{[\{s'\}]} e^{-\beta H'[\{s'\}]} = \sum_{[\{s'\}]}\sum_{[\{s\}]}\prod_{\text{blocks}} T\big(s'; s_1, ..., s_q\big)e^{-\beta H[\{s\}]} \\ &= \sum_{[\{s\}]} e^{-\beta H[\{s\}]}\sum_{[\{s'\}]}\prod_{\text{blocks}} T\big(s'; s_1, ..., s_q\big). \end{aligned} \tag{17}$$

Since $\sum_{s'} T = 1$, one has $Z' = Z$. We can see that the partition function is never changed, we cut integrate out the blocks first and the integrate out the coarse-grained spins later, which is consist with integrating out the highier momentum modes in QFTs.

One may expect a second step of rescaling to restore the original lattice spacing and the Kinetic term to have the field normalized. However, the Ising CFT is independent of the lattice spacing and the field is already normalized in the definition of the BST to have σ fixed to be $+1$ or -1. Thus we do not need to do any rescaling.

From the new Hamiltonian $H'[\{s'\}]$, reading off the new couplings gives an RG map

$$(J', h', ...) = \mathrm{RG}[(J, h, ...)]. \tag{18}$$

1.2.2 One-dimensional decimation

For 1D Ising model, we can do the RG transformation exactly. For $d = 1$ and $h = 0$, choose every two spins to live in a block and integrate out the block with a special BST which only sees the even number blocks and simply ignore the first one. The BST is:

$$T(s'; s_1, s_2) = \delta_{s', s_2}, \qquad H = -J \sum_i s_i s_{i+1} - c \sum_i 1. \tag{19}$$

> **Remark**
>
> Here in the original theory we add a term of $-c$ to the Hamiltonian, which is to nromalize out the vaccum energy.

Summing over inside the blocks gives the coarse-grained Hamiltonian as

$$\begin{aligned} e^{-\beta H'[\{s_{\text{even}}\}]} &= \sum_{s_{\text{odd}}} e^{\beta\left[J\sum_{\text{even}}(s_i s_{i+1} + s_{i+1} s_{i+2}) + 2\sum_{\text{even}} c\right]} \\ &= \prod_{\text{even}} \left[e^{\beta(J(s_i + s_{i+2}) + 2c)} + e^{\beta(-J(s_i + s_{i+2}) + 2c)}\right]. \end{aligned} \tag{20}$$

Notice that here s_i, s_{i+2} are both fixed number. In the final form, the first term is given by $s_{i+1} = 1$ and the second term is given by $s_{i+1} = -1$. Matching this to $e^{J' s_i s_{i+2} + c'}$, and redefining $K = \beta J$, $K' = \beta J'$, and $C = \beta c$, we have:

$$K' = \frac{1}{2}\ln(\cosh(2K)), \qquad C' = 2C + \frac{1}{2}\ln(4\cosh(2K)). \tag{21}$$

With $x = \tanh K$, $0 \le x < 1$,

$$x' = \mathrm{RG}(x) = x^2. \tag{22}$$

Thus the unstable zero-temperature fixed point $x = 1$ flows toward the stable infinite-temperature fixed point $x = 0$.

1.3 Linearized flow near a fixed point

Let K_a be a set of couplings and $K'_a = \mathrm{RG}(K_a)$ be the RG transformation. We define a fixed point K^*_a as a point that is invariant under the RG transformation:

$$K^*_a = \mathrm{RG}(K^*_a). \tag{23}$$

Let K^*_a be a fixed point. Linearizing the RG transformation gives

$$K'_a - K^*_a = \left[\frac{\partial K'_a}{\partial K_b}\right]_{K=K^*} (K_b - K^*_b) \equiv T_{ab}(K_b - K^*_b). \tag{24}$$

Let v^i_a be left eigenvectors (i here is the label of eigenvectors),

$$v^i_a T_{ab} = \lambda_i v^i_b, \tag{25}$$

and define scaling variables

$$u_i = v^i_a (K_a - K^*_a). \tag{26}$$

We now can see that the RG transformation is diagonalized in the u_i basis: Under one RG transformation, the new scaling variable is

$$\begin{aligned} u_i' &= v_a^i (K_a' - K_a^*) \\ &= v_a^i T_{ab} (K_b - K_b^*) + \mathcal{O}\Big((K - K^*)^2\Big) \\ &= \lambda_i v_b^i (K_b - K_b^*) + \mathcal{O}\Big((K - K^*)^2\Big) \\ &= \lambda_i u_i + \mathcal{O}(u^2). \end{aligned} \tag{27}$$

Therefore, to linear order around the fixed point, the RG transformation is diagonal in the scaling-variable basis:

$$u_i' = \lambda_i u_i. \tag{28}$$

After one blocking step, we commonly use b to denote the linear size of the block and use it to characterize how the variable transform as we change the scale.

$$u_i' = \lambda_i u_i \equiv b^{y_i} u_i, \qquad b^d = \text{number of spins per block.} \tag{29}$$

where d is the spatial dimension. Then we can now see how different parameters behave under the RG transformation.

- If $y_i > 0$, then u_i grows under RG and the corresponding operator is relevant.
- If $y_i < 0$, then u_i shrinks under RG and the corresponding operator is irrelevant.
- If $y_i = 0$, then u_i is unchanged under RG and the corresponding operator is marginal. If the deformation of

This is in fact related to the QFT picture, for a operator that perturb the fixed-point action is $\sum_i c_i \mathcal{O}_i$, the scaling dimension of $\mathcal{O}_i$ is:

$$\Delta_i = d - y_i. \tag{30}$$

1.4 Scaling Hypothesis

Most observable in StatMech system can be given from the free energy. Thus the behavior of the free energy under RG transformation near a fixed point is of great interest, it may give us behavior of observables near the critical points in terms of the couplings. The free energy density is defined as:

$$f = -\frac{1}{\beta N} \log Z, \tag{31}$$

Under RG transformation, the partition function is invariant, thus the free energy density transforms as:

$$f' = -\frac{1}{\beta N'} \log Z = b^d f \tag{32}$$

remember that b^d is the number of spins per block. The free energy density can be split into:

$$f(\{K, c\}) = c + \hat{f}(\{K\}), \tag{33}$$

We split out the vacuum energy density c since it is irrelevant to most observables. In general we have:

$$c' = b^d (c + g(\{K\})) \tag{34}$$

Thus under RG transformation, we have the following relation:

$$\hat{f}(\{K\}) = g(\{K\}) + b^{-d} \hat{f}(\{K'\}) \tag{35}$$

We now split the free energy density into singular and regular parts. We mostly care about the singular part since it is the one that gives us the signal of the second order phase transition. Normally, the function $g(\{K\})$ is regular and thus under RG the singular part of the free energy density transforms as:

$$f_s(K) = b^{-d} f_s(K'). \tag{36}$$

Now we focus on the case that we have a fixed point K^* and we are **near the fixed point**. We assume now we only have thermal and magnetic scaling variables u_t and u_h, (Just assume other irrelevant variables goes to 0 near the fixed point), we have:

$$f_s(u_t, u_h) = b^{-d} f_s(b^{y_t} u_t, b^{y_h} u_h). \tag{37}$$

After n iterations,

$$f_s(u_t, u_h) = b^{-nd} f_s(b^{ny_t} u_t, b^{ny_h} u_h). \tag{38}$$

Choose to end on some n so that $|b^{ny_t} u_t| = u_{t0}$. Then plugging in n in terms of other quantities gives:

$$f_s(u_t, u_h) = \left|\frac{u_t}{u_{t0}}\right|^{\frac{d}{y_t}} f_s\left(\operatorname{sgn}(u_t) u_{t0}, u_h \left|\frac{u_t}{u_{t0}}\right|^{-\frac{y_h}{y_t}}\right). \tag{39}$$

Using reduced temperature $t = \frac{T - T_c}{T_c}$ and the relation $u_t = \frac{t}{t_0}, u_h = \frac{h}{h_0}$ (we here won't justify why this is true, one can assume as a fact for Ising), this is commonly written as

$$f_s(t, h) = \left|\frac{t}{t_0}\right|^{\frac{d}{y_t}} \Phi\left(\left(\frac{h}{h_0}\right)\left|\frac{t}{t_0}\right|^{-\frac{y_h}{y_t}}\right). \tag{40}$$

Consequently, taking derivatives gives the behavior of other observables:

$$C \sim \partial_t^2 f_s \sim |t|^{-\alpha}, \quad M \sim \frac{\partial f_s}{\partial h}\,|_{h=0} \sim (-t)^{\beta}, \quad \chi \sim \frac{\partial M}{\partial h}\,|_{h=0} \sim |t|^{-\gamma}, \tag{41}$$

with

$$\alpha = 2 - \frac{d}{y_t}, \qquad \beta = \frac{d - y_h}{y_t}, \qquad \gamma = \frac{2y_h - d}{y_t}. \tag{42}$$

We can see that the critical exponents are determined by y_t and y_h, which are determined by the scaling properties of the coupling near the fixed point. And they are related to each other, cause they are not independent.

In a fixed spatial dimension, the Ising lattice model, uniaxial magnets, the liquid–gas critical point, and one-component scalar φ^4 theory can belong to the same Ising universality class, that flow to the same fixed point at IR and thus have the same critical behavior.

2 Conformal Transformations

We now focus on a theory on the RG fixed point. One can deduce that the RG fixed point definition of the correlation function will lead to the following identity:

$$\langle \mathcal{O}_1(\lambda x_1)...\mathcal{O}_n(\lambda x_n)\rangle = \lambda^{-\Delta_1-...-\Delta_n}\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle. \tag{43}$$

One may find that this is exactly the definition of symmetry under scaling transformations of certain covariant fields. In fact, at RG fixed points we often have a larger symmetry group, the conformal group. In this section we will discuss the conformal group and its implications for correlation functions.

2.1 Conformal transformations

2.1.1 The conformal Killing equation

A conformal transformation is a coordinate transformation $x \to x'$ for which

$$g'_{\mu\nu}(x') = \Omega(x) g_{\mu\nu}(x). \tag{44}$$

> **Remark**
>
> This is a very generalized equation, without having the metric fixed. Starting now, we will fix a metric and use the flat Euclidean metric $\eta_{\mu\nu} = (1, 1, 1, ...)$, which is the identity matrix. Notice that the Minkowski metric will also lead to a similar result.

For an infinitesimal map

$$x'^{\mu} = x^{\mu} + \varepsilon^{\mu}(x), \tag{45}$$

the Jacobian and its inverse are

$$\frac{\partial x'^{\mu}}{\partial x^{\alpha}} = \delta^{\mu}_{\alpha} + \partial_{\alpha}\varepsilon^{\mu}, \qquad \frac{\partial x^{\mu}}{\partial x'^{\alpha}} = \delta^{\mu}_{\alpha} - \partial_{\alpha}\varepsilon^{\mu}. \tag{46}$$

To first order, we have:

$$g'_{\mu\nu} = g_{\mu\nu} - \partial_{\mu}\varepsilon_{\nu} - \partial_{\nu}\varepsilon_{\mu}. \tag{47}$$

Writing $\Omega = 1 - f$, one obtains the conformal Killing equation

$$\partial_{\mu}\varepsilon_{\nu} + \partial_{\nu}\varepsilon_{\mu} = f(x) g_{\mu\nu}, \qquad f(x) = \frac{2}{d}\partial_{\mu}\varepsilon^{\mu}. \tag{48}$$

In flat space, differentiating and permuting indices gives

$$2\partial_{\mu}\partial_{\nu}\varepsilon_{\rho} = \eta_{\rho\nu}\partial_{\mu}f + \eta_{\rho\mu}\partial_{\nu}f - \eta_{\mu\nu}\partial_{\rho}f. \tag{49}$$

Tracing and differentiating imply

$$2\partial^2\varepsilon_{\rho} = (2-d)\partial_{\rho}f, \tag{50}$$

$$(2-d)\partial_{\mu}\partial_{\nu}f = \eta_{\mu\nu}\partial^2 f. \tag{51}$$

Hence

$$\begin{cases} \text{no local constraint on } f & d = 1 \\ \partial^2 f = 0 & d = 2 \\ \partial_{\mu}\partial_{\nu}f = 0 & d > 2. \end{cases} \tag{52}$$

2.1.2 Finite conformal transformations

For $d > 2$, $f = A + B_\mu x^\mu$. We then take the ansatz of ε^μ given as:

$$\varepsilon^\mu = a_\mu + b_{\mu\nu}x^\nu + c_{\mu\nu\rho}x^\nu x^\rho \quad (2.11) \tag{53}$$

Here $c_{\mu\nu\rho} = c_{\mu\rho\nu}$. For now a_μ, $b_{\mu\nu}$, and $c_{\mu\nu\rho}$ are arbitrary constants. Finally, we can plug this form of ε^μ into the conformal killing equation and require the condition to be satisfied for all x. This leads to the following:

- a_μ is free of constraints, and corresponds to infinitesimal translations.
- $b_{\mu\nu} = \alpha\eta_{\mu\nu} + m_{\mu\nu}$, with $m_{\mu\nu} = -m_{\nu\mu}$. The trace part corresponds to an infinitesimal scale transformation. $m_{\mu\nu}$ corresponds to an infinitesimal Lorentz transformation/rotation.
- $c_{\mu\nu\rho} = \eta_{\mu\rho}b_\nu + \eta_{\mu\nu}b_\rho - \eta_{\nu\rho}b_\mu$, for a constant vector b_μ.

The corresponding finite transformations are

$$\begin{aligned} \text{translation:} \quad & x'^\mu = x^\mu + a^\mu, \\ \text{rotation/Lorentz:} \quad & x'^\mu = M^\mu_\nu x^\nu, \\ \text{dilatation:} \quad & x'^\mu = \lambda x^\mu, \\ \text{SCT:} \quad & x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b\cdot x + b^2x^2}. \end{aligned} \tag{54}$$

An important transformation is called inversion is

$$I: \quad x'^\mu = \frac{x^\mu}{x^2}, \tag{55}$$

and an SCT can be written as inversion–translation of $-b$ –inversion.

2.2 Representation

2.2.1 Representation on Functions

Definition 2.1 **Scalar function**

For a scalar function with zero intrinsic weight, we define it as transforming under the following rule with respect to any spacetime transformation $x \to x'$:

$$f'(x') = f(x), \tag{56}$$

Definition 2.2 **Generator**

For a transformation acting on a function, a generator G_a is defined by following with some real parameter ω_a:

$$f'(x) = e^{-i\omega_a G_a} f(x). \tag{57}$$

For a scalar function the generator for arbitrary transformation can be easily derived: We consider an infinitesimal version of a transformation, which is given by:

$$x'^\mu = x^\mu + \omega_a A^\mu_a(x), \qquad A^\mu_a(x) = \frac{\partial x'^\mu}{\partial \omega_a}\,|_{\omega=0}, \tag{58}$$

then we have:

$$\begin{aligned} f'(x) - f(x) &= -i\omega_a G_a f(x) = f(x - \omega_a A_a(x)) - f(x) \\ &= -\omega_a A_a^\mu(x)\partial_\mu f(x). \end{aligned} \tag{59}$$

$$iG_a f(x) = A_a^\mu(x)\partial_\mu f(x). \tag{60}$$

Then with certain conventions, one can easily find the generators of the conformal group on a scalar function:

$$\begin{aligned} P_\mu &= -i\partial_\mu, \\ L_{\mu\nu} &= i\big(x_\mu\partial_\nu - x_\nu\partial_\mu\big), \\ D &= -ix^\mu\partial_\mu, \\ K_\mu &= -i\big(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\big). \end{aligned} \tag{61}$$

Notice that the definition of the generators are convention dependent, however, $\omega_a G_a$ is convention independent. To be more concrete, we choose the convention that ω_a is related to the parameters of the finite conformal transformation as:

$$\omega_a G_a = a^\mu P_\mu + \frac{1}{2}\theta^{\mu\nu}L_{\mu\nu} + \lambda D + b^\mu K_\mu, \qquad \theta^{\mu\nu} = -\theta^{\nu\mu}. \tag{62}$$

The parameters have the following geometric meanings:

- a^μ is the translation distance: $x'^\mu = x^\mu + a^\mu$.
- $\theta^{\mu\nu}$ specifies an infinitesimal rotation in Euclidean spacetime, or a rotation/boost in Minkowski spacetime. Its related to the coordinate transformation by the canonical exponential map:

$$x'^\mu = e^{\frac{1}{2}\theta^{\rho\sigma}L_{\rho\sigma}}x^\mu \quad L_{\rho\sigma} = x_\rho\delta^\mu_\sigma - x_\sigma\delta^\mu_\rho. \tag{63}$$

- λ is the logarithm of the dilation factor: $x'^\mu = e^\lambda x^\mu$.
- b^μ specifies a special conformal transformation. It has the dimension of inverse length. The corresponding finite transformation is

$$x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b_\nu x^\nu + b^2x^2}. \tag{64}$$

Together, these parameters give the infinitesimal coordinate transformation

$$x'^\rho = x^\rho + a^\rho + \theta^{\rho\nu}x_\nu + \lambda x^\rho + 2(b_\nu x^\nu)x^\rho - b^\rho x^2 + O(\omega^2). \tag{65}$$

For example, a finite dilatation gives

$$f'(r') = e^{-i\alpha D}f(r') = e^{-\alpha r'\partial_{r'}}f(r') = f(e^{-\alpha}r') = f(r), \tag{66}$$

where $r' = e^\alpha r$. by exponentialing the derivative we mean it by Taylor expansions.

2.2.2 Conformal algebra

With the conventions above, we can find an algebra of the conformal generators.

$$\begin{aligned} \big[D, P_\mu\big] &= iP_\mu, \quad \big[D, K_\mu\big] = -iK_\mu, \quad \big[K_\mu, P_\nu\big] = 2i\big(\eta_{\mu\nu}D - L_{\mu\nu}\big), \\ \big[L_{\mu\nu}, P_\rho\big] &= -i\big(\eta_{\mu\rho}P_\nu - \eta_{\nu\rho}P_\mu\big), \quad \big[L_{\mu\nu}, K_\rho\big] = -i\big(\eta_{\mu\rho}K_\nu - \eta_{\nu\rho}K_\mu\big), \\ \big[L_{\mu\nu}, L_{\rho\sigma}\big] &= -i\big(\eta_{\mu\rho}L_{\nu\sigma} - \eta_{\mu\sigma}L_{\nu\rho} - \eta_{\nu\rho}L_{\mu\sigma} + \eta_{\nu\sigma}L_{\mu\rho}\big). \\ \big[P_\mu, P_\nu\big] &= \big[K_\mu, K_\nu\big] = [D, D] = \big[D, L_{\mu\nu}\big] = 0, \end{aligned} \tag{67}$$

Defining

$$J_{\mu\nu} = L_{\mu\nu}, \quad J_{-1,\mu} = \frac{1}{2}(P_\mu - K_\mu), \quad J_{0,\mu} = \frac{1}{2}(P_\mu + K_\mu), \quad J_{-1,0} = D, \tag{68}$$

one recognizes the algebra $\mathrm{so}(d+1,1)$ in Euclidean signature. If we consider Minkwoski spacetime with signature $(-,+,+,+,...)$, the algebra is $\mathrm{so}(d,2)$.

> **Remark**
>
> Above discussion is all in Euclidean spacetime, however, one can prove that the Minkowski version is just changing the metric.

> **Remark**
>
> Why functions instead of the defining representation? This question in fact we have encountered in Poincare representation. Its hard to get the commutation of the spacetime translation and the lorentz transformation using the defining representation. Thus we often just use representation on functions or an abstract representation on Hilbert space to get the algebra.

2.2.3 Action on local operators

Now we consider the action of the conformal group on the Hilbert space. For a quantum field theory, the symmetry transformations are represented by unitary operators on the Hilbert space (and generators are Hermite operators, at least for Lorenzian Theory a theory in Euclidean spacetime may be more subtle to consist with the Wick rotation). Consider a field that transforms under some transformation as:

$$x' = F(x), \qquad \varphi'(x') = F(\varphi(x)), \tag{69}$$

> **Definition 2.3** **Quantum generator**
>
> For a quantum operator, it is assumed to be operator (maybe unitary or not) and it act on local fields as:
>
> $$\varphi'(x) = U^{-1}\varphi(x)U, \qquad U = e^{-i\omega_a \hat{G}_a}. \tag{70}$$
>
> Now $\hat{G}_a$ is the quantum generators of the theory and is assumed to be a Hermite operator.

> **Remark**
>
> In lorentzian theory, the requirement of the probability conservation will tell us that the quantum operators should be unitary. However, in Euclidean theory we don't have a notion of time, thus we don't need requirement of unitary. However, we often want the Euclidean theory to have a consist wick rotate lorentzian conterpart. Such requirement will impose conditions on the properties of the quantum operators.

Here we will define ω_a just as the parameters we used for the transformation on scalar functions.

> **Remark**
>
> This relation should be understood as a definition for a quantum generator. In QFT canonical quantization scheme we often find the quantum generators are the conserved charges exactly which lead to this convention.

The infinitesimal action WLOG can contain both an orbital part and a finite-dimensional spin part:

$$\left[\hat{G}_a, \varphi(x)\right] = -G_a\varphi(x) + iR(G_a)\varphi(x). \tag{71}$$

For the momentum generators, we have:

$$\varphi'(x) = e^{ia\cdot\hat{P}}\varphi(x)e^{-ia\cdot\hat{P}}, \quad \varphi'(x) = \varphi(x-a). \quad \varphi(x+a) = e^{-ia\cdot\hat{P}}\varphi(x)e^{ia\cdot\hat{P}}. \tag{72}$$

Thus we have:

$$\varphi(x) = e^{-ix\cdot\hat{P}}\varphi(0)e^{ix\cdot\hat{P}}, \tag{73}$$

> **Remark**
>
> This might look a bit different from the books using $(+,-,-,-)$ conventions, here we are using $(+,+,+,+)$ conventions. This note is self consistent with this convention.

The infinitesimal action of the translation generator is then:

$$\left[\hat{P}_\mu, \varphi(x)\right] = i\partial_\mu\varphi(x). \tag{74}$$

Thus, we may see that we only need to define the action of conformal generators on the local operators at the origin, their commutation relation will tell us how they act on local operators at any points.

2.2.4 Primary and Descendant Fields

Definition 2.4 **Primary Fields**

We define a certain series of local operators φ_α to commute with the generators as:

$$\left[\hat{D}, \varphi_\alpha(0)\right] = i\Delta\varphi_\alpha(0), \quad \left[\hat{L}_{\mu\nu}, \varphi_\alpha(0)\right] = i\left(S_{\mu\nu}\right)_{\alpha\beta}\varphi_\beta(0). \quad \left[K_\mu, \varphi_\alpha(0)\right] = 0. \tag{75}$$

The commutation relations beyond the origin can be derived from the algebra with the translation generator.

$$\begin{aligned}
\left[P_\mu, \varphi_\alpha(x)\right] &= i\partial_\mu\varphi_\alpha(x) \\
\left[D, \varphi_\alpha(x)\right] &= i\left(\Delta + x^\mu\partial_\mu\right)\varphi_\alpha(x) \\
\left[L_{\mu\nu}, \varphi_\alpha(x)\right] &= -i\left(x_\mu\partial_\nu - x_\nu\partial_\mu\right)\varphi_\alpha(x) + i\left(S_{\mu\nu}\right)_{\alpha\beta}\varphi_\beta(x) \\
\left[K_\mu, \varphi_\alpha(x)\right] &= 2ix_\mu\Delta\varphi_\alpha(x) + i\left(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\right)\varphi_\alpha(x) + 2ix^\rho\left(S_{\rho\mu}\right)_{\alpha\beta}\varphi_\beta(x)
\end{aligned} \tag{76}$$

One can also derive the finite transformation of the primary field, for example under the dilation transformation, we have:

$$\varphi'(\lambda x) = e^{i\lambda\hat{D}}\varphi(\lambda x)e^{-i\lambda\hat{D}} = \lambda^{-\Delta}\varphi(x). \tag{77}$$

For a scalar primary field, the finite transformation under a general conformal transformation is:

$$\varphi'(x') = \Omega(x)^{-\frac{\Delta}{d}}\varphi(x). \quad \Omega(x) = \det\frac{\partial x'^\mu}{\partial x^\nu} \tag{78}$$

For a generic scaling field that commute as:

$$[D, \varphi(0)] = i\Delta\varphi(0). \tag{79}$$

The algebra shows that P_μ raises the dimension by one, while K_μ lowers it by one:

$$\begin{aligned}\big[D, P_\mu\varphi(0)\big] &= i(\Delta+1)P_\mu\varphi(0),\\ \big[D, K_\mu\varphi(0)\big] &= i(\Delta-1)K_\mu\varphi(0).\end{aligned} \tag{80}$$

In a unitary theory the spectrum of dimensions is bounded below. Thus we can see that by acting the P_μ operators on a primary field we can generate a seires of scaling field with highier scaling dimensions:

$$\big[P_{\mu_1}...P_{\mu_n}, \varphi(0)\big] \sim \partial_{\mu_1}...\partial_{\mu_n}\varphi(0). \tag{81}$$

We call these fields descendants of the primary field.

2.3 Noether's theorem and the stress tensor

For an action

$$S = \int d^dx\mathcal{L}\big(\varphi, \partial_\mu\varphi\big), \tag{82}$$

Noether's theorem associates a conserved current to every continuous symmetry. In a general infinitesimal transformation, one convenient form is

$$j^\mu_a = \left[\frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\partial_\nu\varphi - \delta^\mu_\nu\mathcal{L}\right]\frac{\partial x^\nu}{\partial\omega_a} - \frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\frac{\partial\tilde{F}}{\partial\omega_a}, \qquad \partial_\mu j^\mu_a = 0. \tag{83}$$

The translation current is the canonical stress tensor,

$$T^{\mu\nu}_{\mathrm{c}} = -\eta^{\mu\nu}\mathcal{L} + \frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\partial^\nu\varphi, \qquad \partial_\mu T^{\mu\nu}_{\mathrm{c}} = 0. \tag{84}$$

By an improvement one may choose a specific symmetric tensor $T^{\mu\nu} = T^{\nu\mu}$. Under a general coordinate transformation $x'^\mu = x^\mu + \varepsilon^\mu(x)$, we have:

$$\delta S = -\frac{1}{2}\int d^dx T^{\mu\nu}\big(\partial_\mu\varepsilon_\nu + \partial_\nu\varepsilon_\mu\big) \tag{85}$$

If we choose this general transformation to be a conformal transformation, we have due to the conformal killing equation:

$$\delta S = -\frac{1}{2}\int d^dx f(x) T^\mu{}_\mu \tag{86}$$

Scale invariance means that $\delta S = 0$ when $f(x) = \mathrm{const}$ this will lead to:

$$T^\mu{}_\mu = \partial_\mu V^\mu, \tag{87}$$

with a virial current V^μ. If we also have conformal invariance, we have $\delta S = 0$ for SCT, then:

$$T^\mu{}_\mu = 0. \tag{88}$$

In summary, in CFT we always can have a improved symmetric and traceless stress tensor $T^{\mu\nu}$, which is conserved.

2.4 Implications for correlation functions

For a QFT we define the correlation function as:

$$\langle\mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle = \left(\frac{1}{Z}\right)\int D\varphi\mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)e^{-S[\varphi]}, \tag{89}$$

notice that we always work in the Euclidean signature. One can prove that for a symmetry we have the result that:

$$\langle\varphi(x_1')...\varphi(x_n')\rangle = \langle\mathcal{F}(\varphi(x_1))...\mathcal{F}(\varphi(x_n))\rangle \tag{90}$$

By $\mathcal{F}(\varphi(x))$ we means the RHS of the definition of field condiguration transformation:

$$\varphi'(x') = \mathcal{F}(\varphi(x)). \tag{91}$$

which is the transformation purely of field (not the spacetime point). Notice that this is pure path integral and have nothing to do with whether the transformation is Unitary in certain time choice or not.

2.4.1 Two-point functions

Translation and rotation invariance imply dependence only on $x_{12} = x_1 - x_2$. Dilatations give

$$\langle\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\rangle = \frac{C_{12}}{(|x_{12}|)^{\Delta_1+\Delta_2}}. \tag{92}$$

Under an SCT,

$$x_{12'}^2 = \frac{x_{12}^2}{\gamma_1\gamma_2}, \qquad \gamma_i = 1 - 2b\cdot x_i + b^2 x_i^2. \tag{93}$$

SCT covariance requires $\Delta_1 = \Delta_2$. In an orthonormal basis of scalar primaries,

$$\langle\mathcal{O}_i(x_1)\mathcal{O}_j(x_2)\rangle = \frac{C_i\delta_{ij}}{(x_{12}^2)^{\Delta_i}}. \tag{94}$$

2.4.2 Three-point functions

For scalar primaries, conformal symmetry fixes the position dependence completely:

$$\langle\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = \frac{\lambda_{123}}{\left[(x_{12}^2)^{\frac{\Delta_1+\Delta_2-\Delta_3}{2}}(x_{13}^2)^{\frac{\Delta_1+\Delta_3-\Delta_2}{2}}(x_{23}^2)^{\frac{\Delta_2+\Delta_3-\Delta_1}{2}}\right]}. \tag{95}$$

2.4.3 Four-point functions and cross-ratios

For four identical scalar primaries of dimension Δ,

$$\langle\mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \frac{1}{\left[(x_{12}^2)^{\Delta}(x_{34}^2)^{\Delta}\right]}, \mathcal{G}(u,v), \tag{96}$$

where

$$u = \frac{x_{12}^2x_{34}^2}{x_{13}^2x_{24}^2} = z\overline{z}, \qquad v = \frac{x_{14}^2x_{23}^2}{x_{13}^2x_{24}^2} = (1-z)(1-\overline{z}). \tag{97}$$

Notice that the definition of z and $\overline{z}$ is conventionally not the same for Euclidean and Lorentzian theory (though the corrator will always take the same form). One can see that in Euclidean signature $\overline{z} = z^*$. In Lorentzian signature z and $\overline{z}$ are independent real variables in the standard kinematic region due to the contraction definition difference.

A standard Lorentzian configuration is

$$x_1 = 0, \qquad x_2 = (t, x, 0, ...), \qquad x_3 = (0, 1, 0, ...), \qquad x_4 = \infty. \tag{98}$$

One should be careful with what we mean by $x_4 = \infty$. In fact, we can take a limit of x_4 to infinity, and the correlation function will be defined as:

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \lim_{x_4\to\infty} (x_4^2)^{\Delta} \langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle. \tag{99}$$

Then

$$u = x^2 - t^2, \qquad v = (1-x)^2 - t^2, \qquad z = x - t, \qquad \overline{z} = x + t. \tag{100}$$

We can explicitly see that in the case z and $\overline{z}$ are independent real variables.

3 Radial Quantization

In the last discussion, we focus on a general notation that is better suited for lorentzian signature. In the Euclidean theory (especially in Radial quantization we will discuss), the Unitarity of the operators is not satisfied. A corresponding concept is the reflection positivity, one can see more on this discussion in Simmons-Duffin's lecture. A brief reason is that we want the wick rotation to be consistent, and the Lorentzian theory is unitary, the Euclidean theory will be reflection positive yet the Euclidean transformation is not unitarily represented.

However, we still want in the certain Hilbert space the time generators is Hermitian. Thus, we conventionally define the generators in another exponential map:

$$O = e^{-i\omega_a G_a} = e^{\omega_a G_a^E}, \tag{101}$$

$$P_\mu{}^E = -iP_\mu, \quad K_\mu{}^E = -iK_\mu, \quad M_{\mu\nu}{}^E = -iM_{\mu\nu}, \quad D^E = -iD. \tag{102}$$

correspondingly, the generators O will be:

$$\omega_a G_a^E = a_\mu P_\mu{}^E + b_\mu K_\mu{}^E + \frac{1}{2}\theta_{\mu\nu} M_{\mu\nu}{}^E + \lambda D^E. \tag{103}$$

We will see that the dilatation generators will be Hermitian in the radial quantization Hilbert space. From now on we will drop the superscript E and use the new convention for the generators. And the commutation relations between the generators and the local primaries will be changed into:

$$[D, \varphi_\alpha(0)] = \Delta\varphi_\alpha(0), \quad \big[M_{\mu\nu}, \varphi_\alpha(0)\big] = \big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(0), \quad \big[K_\mu, \varphi_\alpha(0)\big] = 0. \tag{104}$$

The action of the conformal generators at a general spacetime point can be derived using translations:

$$\begin{aligned}
\big[P_\mu, \varphi_\alpha(x)\big] &= \partial_\mu\varphi_\alpha(x), \\
[D, \varphi_\alpha(x)] &= \big(\Delta + x^\mu\partial_\mu\big)\varphi_\alpha(x), \\
\big[M_{\mu\nu}, \varphi_\alpha(x)\big] &= -\big(x_\mu\partial_\nu - x_\nu\partial_\mu\big)\varphi_\alpha(x) + \big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(x), \\
\big[K_\mu, \varphi_\alpha(x)\big] &= 2x_\mu\Delta\varphi_\alpha(x) + \big(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\big)\varphi_\alpha(x) + 2x^\rho\big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(x).
\end{aligned} \tag{105}$$

In this convention, the Euclidean conformal algebra is

$$\begin{aligned}
\big[D, P_\mu\big] &= P_\mu, \quad \big[D, K_\mu\big] = -K_\mu, \\
\big[K_\mu, P_\nu\big] &= 2\big(\delta_{\mu\nu}D - M_{\mu\nu}\big), \quad \big[M_{\mu\nu}, P_\rho\big] = -\big(\delta_{\mu\rho}P_\nu - \delta_{\nu\rho}P_\mu\big), \\
\big[M_{\mu\nu}, K_\rho\big] &= -\big(\delta_{\mu\rho}K_\nu - \delta_{\nu\rho}K_\mu\big), \\
\big[M_{\mu\nu}, M_{\rho\sigma}\big] &= -\delta_{\mu\rho}M_{\nu\sigma} + \delta_{\mu\sigma}M_{\nu\rho} + \delta_{\nu\rho}M_{\mu\sigma} - \delta_{\nu\sigma}M_{\mu\rho}, \\
\big[P_\mu, P_\nu\big] &= 0, \quad \big[K_\mu, K_\nu\big] = 0, \quad [D, D] = 0, \quad \big[D, M_{\mu\nu}\big] = 0.
\end{aligned} \tag{106}$$

Translations reconstruct an operator away from the origin is given by:

$$\mathcal{O}(x) = e^{x^\mu P_\mu}\mathcal{O}(0)e^{-x^\mu P_\mu}. \tag{107}$$

In this convention the descendants are defined as:

$$P_{\mu_1}...P_{\mu_n}\varphi_\alpha(0). \tag{108}$$

3.1 Radial Quantization

3.1.1 Lorentzian Theory in Cylinder Coordinate

Write a point in $\mathbb{R}^d$ as the following coordinates (τ, n):

$$x^\mu = rn^\mu, \quad r = e^\tau, \quad n^\mu n_\mu = 1. \tag{109}$$

One can verify that this is indeed a conformal transformation. For a **scalar primary** of dimension Δ, we define its form on the cylinder coordinates as:

$$\mathcal{O}_{\text{cyl}}(\tau, n) = e^{\tau\Delta}\mathcal{O}_{\text{flat}}(e^\tau, n). \tag{110}$$

The origin of the plane corresponds to $\tau \to -\infty$, while spatial infinity corresponds to $\tau \to +\infty$. From this relation, we can see how the conformal generators act on the cylinder coordinates. The commutation relations between the conformal generators and the cylinder coordinate primaries are given by:

$$\nabla_\mu{}^{\text{S}} := \big(\delta_\mu{}^\nu - n_\mu n^\nu\big)\frac{\partial}{\partial n^\nu}. \tag{111}$$

The conformal generators act on the cylinder operator as

$$\begin{aligned}
\big[D, \mathcal{O}_{\text{cyl}}(\tau, n)\big] &= \partial_\tau \mathcal{O}_{\text{cyl}}(\tau, n), \\
\big[M_{\mu\nu}, \mathcal{O}_{\text{cyl}}(\tau, n)\big] &= -\Big(n_\mu \nabla_\nu{}^{\text{S}} - n_\nu \nabla_\mu{}^{\text{S}}\Big)\mathcal{O}_{\text{cyl}}(\tau, n), \\
\big[P_\mu, \mathcal{O}_{\text{cyl}}(\tau, n)\big] &= e^{-\tau}\Big(n_\mu(\partial^\tau - \Delta) + \nabla_\mu{}^{\text{S}}\Big)\mathcal{O}_{\text{cyl}}(\tau, n), \\
\big[K_\mu, \mathcal{O}_{\text{cyl}}(\tau, n)\big] &= e^{\tau}\Big(n_\mu(\partial^\tau + \Delta) - \nabla_\mu{}^{\text{S}}\Big)\mathcal{O}_{\text{cyl}}(\tau, n).
\end{aligned} \tag{112}$$

We care about the cylinder coordinate because for a CFT in canonical formalism, we usually consider a Lorentzian CFT on a cylinder, we do the canonical quantization for lorentzian theory to get the Hilbert space. By wick rotation and coordinate symmetry transformation, we can investigate this theory in Euclidean cylinder, and with a coordinate transformation above we can have a theory on the plane.

> **Remark**
>
> Notice that we only do quantization in the lorentzian theory, and the Hilbert space sstays the same no matter what we do to the cooridnate transformation, its just symmetry transformations on the same Hilbert space.

From this procedure, the canonical Hamiltonian of the lorentzian theory (which is definitely a Hermite operator in the Hilbert space) which we quantized on will be something like:

$$U(t) = e^{-itH}, \quad H^\dagger = H. \quad \varphi(t, x) = U(t)^\dagger \varphi(0, x) U(t). \tag{113}$$

under the wick rotation, we have $t \to -i\tau$, and the Euclidean time evolution operator will be:

$$U(\tau) = e^{-\tau H}, \quad H^\dagger = H. \quad \varphi(\tau, x) = e^{\tau H}\varphi(0, x)e^{-\tau H}. \tag{114}$$

Notice that here the Hamiltonian is still Hermitian in the Hilbert space, but the Eculidean time evolution operator is not unitary. Taking a derivative on both side, this gives the commutation relation between the Hamiltonian and the local operators:

$$[H, \varphi(\tau, x)] = \partial_\tau \varphi(\tau, x). \tag{115}$$

One may notice that this is exactly what we called the dilatation operator D. Thus, we identify the dilatation operator (which is a abstract generator) with the Hamiltonian of the lorentzian theory on the cylinder, which is a Hermitian operator on the Hilbert space.

The rotation on the equal time slice in the original lorentzian theory will be generated by:

$$U(R) = e^{-i\frac{1}{2}\theta M^C_{\mu\nu}}, \quad \left(M^C_{\mu\nu}\right)^\dagger = M^C_{\mu\nu}. \tag{116}$$

However in our convention, the generator of the rotation is given by :

$$U(R) = e^{\frac{1}{2}\theta M_{\mu\nu}}, \quad M^\dagger_{\mu\nu} = -M_{\mu\nu}. \tag{117}$$

Thus, one may identify that

$$M_{\mu\nu} = -iM^C_{\mu\nu}. \tag{118}$$

However, due to we insist that the lorentzian theory is Unitary, the rotation generator is anti-Hermite in the Hilbert space.

3.1.2 Radial Quantization

Now we are convinced that it is consistent to define a canonical formalism of the CFT on the Euclidean plane as the one that quantized on the lorentzian theory on the Cylinder (which is well defined in ordinary QFT courses). However, it is too complicated to first do a quantization on a cylinder in lorentzian signature and then wick rotate to the Euclidean plane. Thus, what we do is that we develop a formalism purely on the plane but gives the exactly same Hilbert space as the Unitary lorentzian theory on the cylinder. This approach is called the **radial quantization**.

We take the following conditions for a Unitary Hilbert space:

- D is a Hermite Operator in the Hilbert space and is bounded from below.
- We exist a vacuum state $|0\rangle$ which is invariant under the conformal group.

For a state $|\psi\rangle$ in the Hilbert space, we can act n times with K_μ on it and due to D is bounded from below, we will eventually get a series of state $|\psi_0\rangle$ which is annihilated by all K_μ. Thus the Hilbert space should be spanned by the states of the form:

$$P_{\mu_1}...P_{\mu_n}|\psi_0\rangle. \tag{119}$$

We then can conclude the Hilbert space should be spanned by a series of primary states and its descendants. The primary states satisfies the following conditions:

$$K_\mu|\psi_\alpha\rangle = 0, \quad D|\psi_\alpha\rangle = \Delta|\psi_\alpha\rangle, \quad M_{\mu\nu}|\psi_\alpha\rangle = \left(S_{\mu\nu}\right)_{\alpha\beta}|\psi_\beta\rangle. \tag{120}$$

we then define a primary operator in this Hilbert space as:

Definition 3.1 **Primary Operator**

A primary operator correpsond to the state $|O_i\rangle$ is defined as a local operator O_i such that:

$$O_i(0)|0\rangle = |O_i\rangle, \tag{121}$$

Remark

This relation is more of a axiom that such operator exists not merely if it exist we can do such identification.

Then one can find that these operator indeed have the correct commutation relation with the conformal generators as the ordinary conformal primary fields. More importantly, this construction tells us that a CFT should have a one-to-one correspondence between the primary operators and the primary states in the Hilbert space. This is called the **state-operator correspondence**.

YL: [The discussion here deserves more rigorous consideration. However, I believe all these statements are arguments the true rigorous statement should be discussed in axiomatic approach to CFT not just arguing what a universal behavior should be like, we can't derive this without sufficient axioms.]

3.1.3 Correlation Functions

Before we only define the correlation function in the path integral formalism. Now with a canonical formalism, we need to find an equivalent way to compute the same object. We find out that the correlation function can be computed in the Hilbert space as:

$$\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle = \langle 0|\mathcal{R}\mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)|0\rangle. \tag{122}$$

The LHS is the correlation function defined in the path integral formalism, and the RHS is a correlator in the Hilbert space. (Of course, they should consistent with the same action S)

YL: [This certainly needs proof. However, one can convince themselves this is true.]

3.1.4 Radial Conjugation

For a scalar primary, a consistent radial conjugation (The complex conjugation of radial quantization) acts as inversion:

$$(\mathcal{O}(x))^\dagger = \frac{1}{|x|^{2\Delta}}\mathcal{O}\left(\frac{x}{x^2}\right). \tag{123}$$

The bra corresponding to $|\mathcal{O}\rangle$ is therefore

$$\langle \mathcal{O}| = \lim_{|x|\to\infty} |x|^{2\Delta}\langle 0|\mathcal{O}(x). \tag{124}$$

If the two-point function is normalized by

$$\langle \mathcal{O}(x)\mathcal{O}(0)\rangle = \frac{1}{|x|^{2\Delta}}, \tag{125}$$

then

$$\langle \mathcal{O}|\mathcal{O}\rangle = \lim_{|x|\to\infty} |x|^{2\Delta}\langle \mathcal{O}(x)\mathcal{O}(0)\rangle = 1. \tag{126}$$

The adjoint relations between conformal generators within this Hermite conjugation are:

$$D^\dagger = D, \quad {P_\mu}^\dagger = K_\mu \quad D^\dagger = D, \quad M_{\mu\nu}^\dagger = -M_{\mu\nu}. \tag{127}$$

These relations are the key input in the derivation of the unitarity bounds below.

3.2 Operator product expansion

Consider two local operators φ_1 and φ_2 inserted at points x and 0. In radial quantization, the state created by these two operators is:

$$\varphi_1(x)\varphi_2(0)|0\rangle. \tag{128}$$

Expanding this state in a complete basis of dilatation eigenstates gives a convergent sum over conformal multiplets:

$$\varphi_1(x)\varphi_2(0)|0\rangle = \sum_i C_i(x,\partial)\mathcal{O}_i(0)|0\rangle. \tag{129}$$

Equivalently, inside correlation functions, we can formally write a relation:

$$\varphi_1(x)\varphi_2(0) = \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} C_{\mathcal{O}}(x,\partial)\mathcal{O}(0). \tag{130}$$

The sum runs over primary operators; the differential operator $C_{\mathcal{O}}(x,\partial)$ generates the complete tower of descendants. Conformal symmetry fixes this differential operator in terms of the dimensions and spins, up to the theory-dependent OPE coefficient $\lambda_{12\mathcal{O}}$.

> **Remark**
>
> In fact OPE should be defined within the Radial ordering of the operators. Here it is implicit cause we choose x and 0 points.

3.2.1 Leading terms and the first descendant

For a scalar primary $\mathcal{O}$ of dimension Δ appearing in the OPE of scalar primaries φ_1 and φ_2, the leading terms are

$$\varphi_1(x)\varphi_2(0) \supseteq \frac{\lambda_{12\mathcal{O}}}{|x|^k}\big(\mathcal{O}(0) + \alpha x^\mu \partial_\mu \mathcal{O}(0) + ...\big). \tag{131}$$

We can match the coefficients by the following approach. We compare the action of D, K_μ on the OPE. However, one way is that we first do an OPE expand to the RHS and then act with the generators. The other way is that we first act with the generators on the LHS and then do an OPE expand to the RHS. The two results should be the same. Comparing the terms should fix the coefficients.

Acting with D fixes the power of $|x|$:

$$k = \Delta_1 + \Delta_2 - \Delta. \tag{132}$$

Acting with the special conformal generator fixes the relative coefficient of the first descendant. For a scalar primary,

$$\big[K_\mu, \mathcal{O}(x)\big] = \big(2\Delta x_\mu + 2x_\mu x^\nu \partial_\nu - x^2\partial_\mu\big)\mathcal{O}(x). \tag{133}$$

The result is

$$\alpha = \frac{\Delta_1 + \Delta - \Delta_2}{2\Delta}. \tag{134}$$

In particular, the numerator contains $\Delta_1 - \Delta_2$, not $\Delta_1 + \Delta_2$. The identity contribution $\Delta = 0$ must be treated separately and has no descendants.

The same coefficient can be obtained directly by inserting the OPE into a three-point function. Conformal invariance fixes

$$\langle\varphi_1(x_1)\varphi_2(x_2)\mathcal{O}(x_3)\rangle = \frac{\lambda_{12\mathcal{O}}}{|x_{12}|^{\Delta_1+\Delta_2-\Delta}|x_{23}|^{\Delta_2+\Delta-\Delta_1}|x_{13}|^{\Delta_1+\Delta-\Delta_2}}. \tag{135}$$

And pluging in the OPE ansatz into the 3 point correlation function for the first two fields and then use the equivalent of the canonical and path integral formalism tells us that :

$$\begin{aligned}\langle\varphi_1(x)\varphi_2(0)\mathcal{O}(x_3)\rangle &= \langle 0|\mathcal{R}\varphi_1(x)\varphi_2(0)\mathcal{O}(x_3)|0\rangle \\ &= \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} C_{\mathcal{O}}\big(x_{12},\partial_{x_2}\big)\langle\mathcal{O}(x_2)\mathcal{O}(x_3)\rangle|_{x_2=0}.\end{aligned} \tag{136}$$

3.3 Unitarity bounds

In radial quantization, positivity of descendant norms constrains the possible scaling dimensions. For a scalar primary state $|\mathcal{O}\rangle$,

$$\big\langle\mathcal{O}|K_\mu P_\nu|\mathcal{O}\big\rangle = 2\Delta\delta_{\mu\nu}\langle\mathcal{O}|\mathcal{O}\rangle. \tag{137}$$

Thus positivity at level one implies $\Delta \geq 0$. The stronger scalar bound comes from the level-two descendant

$$P^\mu P_\mu|\mathcal{O}\rangle. \tag{138}$$

Its norm is

$$\big\|P^\mu P_\mu|\mathcal{O}\rangle\big\|^2 = 8d\Delta\left(\Delta - \frac{d-2}{2}\right)\||\mathcal{O}\rangle\|^2. \tag{139}$$

Therefore a scalar primary in a unitary CFT satisfies

$$\Delta = 0 \quad \text{for the identity,} \quad \text{or} \quad \Delta \geq \frac{d-2}{2}. \tag{140}$$

For a symmetric traceless primary of spin $\ell \geq 1$,

$$\mathcal{O}_{\mu_1\ldots\mu_\ell}(x), \tag{141}$$

positivity of the level-one divergence descendant gives

$$\Delta \geq d - 2 + \ell. \tag{142}$$

3.4 Conformal Blocks

For four identical scalar primaries of dimension Δ_φ, conformal symmetry gives

$$\langle\varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4)\rangle = \frac{\mathcal{G}(u,v)}{|x_{12}|^{2\Delta_\varphi}|x_{34}|^{2\Delta_\varphi}}, \tag{143}$$

where the conformal cross-ratios are

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \quad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}. \tag{144}$$

Applying the OPE to the pairs (12) and (34) yields the conformal-block decomposition

$$\mathcal{G}(u,v) = \sum_{\mathcal{O}} \lambda^2_{\varphi\varphi\mathcal{O}} G_{\Delta,\ell}(u,v). \tag{145}$$

More explicitly, we have:

$$\Big[C_{\Delta(x_{12},\partial_y)} C_{\Delta(x_{34},\partial_z)}\langle\varphi_\Delta(y)\varphi_\Delta(z)\rangle\Big]\ |_{y=\frac{x_1+x_2}{2},z=\frac{x_3+x_4}{2}} = \frac{G_{\Delta,\ell}(u,v)}{|x_{12}|^{2\Delta_E}|x_{34}|^{2\Delta_E}} \tag{146}$$

The conformal block $G_{\Delta,\ell}(u,v)$ contains the contribution of one exchanged primary of dimension Δ and spin ℓ, together with all of its descendants. Once the external dimensions and the quantum numbers (Δ,ℓ) are specified, the block is fixed entirely by conformal symmetry.It solves the Casimir differential equation with appropriate boundary conditions.

4 2D Conformal Field Theory

4.1 2D Conformal Algebra

Recall the conformal Killing equation for $x'^{\mu} = x^{\mu} + \varepsilon^{\mu}(x)$:

$$\partial_{\mu}\varepsilon_{\nu} + \partial_{\nu}\varepsilon_{\mu} = \frac{2}{d}(\partial_{\rho}\varepsilon^{\rho})\eta_{\mu\nu}. \tag{147}$$

In two Euclidean dimensions this gives

$$\partial_0\varepsilon_0 = \partial_1\varepsilon_1, \quad \partial_0\varepsilon_1 = -\partial_1\varepsilon_0. \tag{148}$$

Let

$$z = x^0 + ix^1, \quad \overline{z} = x^0 - ix^1, \tag{149}$$

$$\partial = \partial_z = \frac{1}{2}(\partial_0 - i\partial_1), \quad \overline{\partial} = \partial_{\overline{z}} = \frac{1}{2}(\partial_0 + i\partial_1). \tag{150}$$

The nonzero metric components are

$$g_{z\overline{z}} = g_{\overline{z}z} = \frac{1}{2}, \quad g^{z\overline{z}} = g^{\overline{z}z} = 2. \tag{151}$$

Writing $\varepsilon = \varepsilon^0 + i\varepsilon^1$ and $\overline{\varepsilon} = \varepsilon^0 - i\varepsilon^1$, we obtain

$$\overline{\partial}\varepsilon = 0, \quad \partial\overline{\varepsilon} = 0. \tag{152}$$

Hence $\varepsilon = \varepsilon(z)$ is holomorphic and $\overline{\varepsilon} = \overline{\varepsilon}(\overline{z})$ is antiholomorphic. Finite conformal coordinate transformations have the form

$$z' = f(z), \quad \overline{z}' = \overline{f}(\overline{z}). \tag{153}$$

Expand the infinitesimal transformation as

$$\varepsilon(z) = \sum_n c_n z^{n+1}, \quad \overline{\varepsilon}(\overline{z}) = \sum_n \overline{c}_n \overline{z}^{n+1}. \tag{154}$$

For a scalar function, the induced variation at fixed coordinates is

$$\delta_{\varepsilon}F(z,\overline{z}) = F(z - \varepsilon(z), \overline{z} - \overline{\varepsilon}(\overline{z})) - F(z,\overline{z}) = -\varepsilon(z)\partial F - \overline{\varepsilon}(\overline{z})\overline{\partial}F. \tag{155}$$

Thus we introduce the generators

$$\ell_n = -z^{n+1}\partial, \quad \overline{\ell}_n = -\overline{z}^{n+1}\overline{\partial}. \tag{156}$$

They form two commuting copies of the Witt algebra:

$$[\ell_m, \ell_n] = (m-n)\ell_{m+n}, \quad \left[\overline{\ell}_m, \overline{\ell}_n\right] = (m-n)\overline{\ell}_{m+n}, \quad \left[\ell_m, \overline{\ell}_n\right] = 0. \tag{157}$$

Consider the Riemann sphere S^2, with coordinate charts z and $z' = -\frac{1}{z}$. The vector field ℓ_n is regular at $z = 0$ only if $n \geq -1$, and regular at $z' = 0$ only if $n \leq 1$. Hence the globally defined holomorphic generators are $\ell_{-1}, \ell_0, \ell_1$, together with their antiholomorphic counterparts. They form the global conformal algebra. The combination $\ell_0 + \overline{\ell}_0$ generates dilatations, while $i\big(\ell_0 - \overline{\ell}_0\big)$ generates rotations.

Finite global transformations are Möbius maps:

$$z \to \frac{az+b}{cz+d}, \quad ad - bc = 1. \tag{158}$$

Their group is $PSL(2,\mathbb{C}) = \frac{SL(2,\mathbb{C})}{\{+1,-1\}} \cong SO^+(3,1)$.

Define a two-dimensional conformal primary by its transformation under every local conformal map:

Definition 4.1 **Primary Field**

A primary field $\varphi(z,\overline{z})$ of weights $\left(h,\overline{h}\right)$ transforms as:

$$\varphi'(z',\overline{z}') = \left(\frac{\partial z'}{\partial z}\right)^{-h}\left(\frac{\partial \overline{z}'}{\partial \overline{z}}\right)^{-\overline{h}}\varphi(z,\overline{z}). \tag{159}$$

A quasiprimary obeys this law under global conformal transformations only. Its scaling dimension and spin are defined as:

$$\Delta = h+\overline{h}, \quad s = h-\overline{h}. \tag{160}$$

Remark

Notice that here we are not only dealing with scalar primaries, but also fields with spin. Due to we are working in 2D, spin are only represented by a single number and U(1) phase. In general the spin is only integer or half-integer.

For a primary field, the infinitesimal transformation law is:

$$\delta_{\varepsilon,\overline{\varepsilon}}\varphi(z,\overline{z}) = -(h\partial\varepsilon+\varepsilon\partial)\varphi(z,\overline{z}) - \left(\overline{h}\overline{\partial}\overline{\varepsilon}+\overline{\varepsilon}\overline{\partial}\right)\varphi(z,\overline{z}). \tag{161}$$

Sometimes we separate the holomorphic and antiholomorphic parts, and give the notation of:

$$\delta_{\varepsilon}\varphi(z,\overline{z}) = -(h\partial\varepsilon+\varepsilon\partial)\varphi(z,\overline{z}), \quad \delta_{\overline{\varepsilon}}\varphi(z,\overline{z}) = -\left(\overline{h}\overline{\partial}\overline{\varepsilon}+\overline{\varepsilon}\overline{\partial}\right)\varphi(z,\overline{z}). \tag{162}$$

However, one should remember that only the sum of the two variations is the actual infinitesimal conformal transformation of the field. For quasi-primary we have the same transformation under global conformal transformations, but not under local ones.

4.2 Holomorphic stress tensor

Symmetry and tracelessness imply

$$g^{\mu\nu}T_{\mu\nu} = 2(T_{z\overline{z}}+T_{\overline{z}z}) = 0, \tag{163}$$

so $T_{z\overline{z}}=0$. Conservation then gives

$$\partial^{\mu}T_{\mu\nu} = 0 \quad \Rightarrow \quad \overline{\partial}T_{zz} = 0, \quad \partial T_{\overline{z}\overline{z}} = 0. \tag{164}$$

Define the holomorphic and antiholomorphic components by

$$T(z) = -2\pi T_{zz}, \quad \overline{T}(\overline{z}) = -2\pi T_{\overline{z}\overline{z}}. \tag{165}$$

4.3 Conformal correlators

YL: [stopped here.]

These formulas also hold for quasiprimaries. For primaries of weights $\left(h_i,\overline{h}_i\right)$,

$$\left\langle\varphi_i(z_1,\overline{z}_1)\varphi_j(z_2,\overline{z}_2)\right\rangle = \frac{C_{ij}}{z_{12}^{2h_i}\overline{z}_{12}^{2\overline{h}_i}}. \tag{166}$$

The coefficient can be nonzero only when the weights match. The three-point function is

$$\begin{aligned}&\left\langle\varphi_i(z_1,\overline{z}_1)\varphi_j(z_2,\overline{z}_2)\varphi_k(z_3,\overline{z}_3)\right\rangle\\&=\frac{\lambda_{ijk}}{z_{12}^{h_i+h_j-h_k}z_{13}^{h_i+h_k-h_j}z_{23}^{h_j+h_k-h_i}\overline{z}_{12}^{\overline{h}_i+\overline{h}_j-\overline{h}_k}\overline{z}_{13}^{\overline{h}_i+\overline{h}_k-\overline{h}_j}\overline{z}_{23}^{\overline{h}_j+\overline{h}_k-\overline{h}_i}}.\end{aligned}\tag{167}$$

Four-point functions are more complicated. For identical holomorphic fields, their coordinate dependence is a fixed prefactor times a function of the cross ratio

$$\eta=\frac{z_{12}z_{34}}{z_{13}z_{24}}.\tag{168}$$

A general form of the 4 point function is:

$$\langle\varphi(z_1,\overline{z}_1)\varphi(z_2,\overline{z}_2)\varphi(z_3,\overline{z}_3)\varphi(z_4,\overline{z}_4)\rangle=\frac{G(\eta,\overline{\eta})}{z_{12}^{2h}z_{34}^{2h}\overline{z}_{12}^{2\overline{h}}\overline{z}_{34}^{2\overline{h}}}.\tag{169}$$

where $G(\eta,\overline{\eta})$ is some function that only depends on the cross ratio.

4.4 Radial quantisation and the OPE

One may find that the conformal transformations are generated by the following charges. In our conventions,

$$Q=\frac{1}{2\pi i}\oint\left(\mathrm{d}z\varepsilon(z)T(z)+\mathrm{d}\overline{z}\overline{\varepsilon}(\overline{z})\overline{T}(\overline{z})\right).\tag{170}$$

> **Remark**
>
> In the yellow book, we prove the conformal ward identity in the path integral formalism and this relation can be derived from it and the correspondence between the path integral and operator formalism. Here we understand as just stating a consistent result.

For a holomorphic variation, one may find that:

$$\delta_\varepsilon\psi(w)=-\frac{1}{2\pi i}\left[\oint_C\mathrm{d}z\varepsilon(z)T(z),\psi(w)\right],\tag{171}$$

The countour should be a equal time circle.

where the commutator implements radial ordering. This is analogous to time ordering in Lorentzian QFT. In radial quantisation,

$$\langle\varphi(z_1,\overline{z}_1)...\varphi(z_n,\overline{z}_n)\rangle=\langle0|R[\varphi(z_1,\overline{z}_1)...\varphi(z_n,\overline{z}_n)]|0\rangle.\tag{172}$$

If $|w_1|<...<|w_n|$, radial ordering places the operator at w_n furthest to the left and the operator at w_1 furthest to the right.

For holomorphic $a(z)$ and another field $b(w)$, a contour just outside $|w|$ minus one just inside $|w|$ gives

$$\oint_{C_w}\mathrm{d}zR[a(z)b(w)]=\oint_{C_+}\mathrm{d}za(z)b(w)-\oint_{C_-}\mathrm{d}zb(w)a(z)=\left[\oint_{|z|=|w|}\mathrm{d}za(z),b(w)\right].\tag{173}$$

The infinitesimal transformation law of a primary is

$$\delta_{\varepsilon,\overline{\varepsilon}}\varphi(z,\overline{z})=-(h\partial\varepsilon+\varepsilon\partial)\varphi(z,\overline{z})-\left(\overline{h\partial}\overline{\varepsilon}+\overline{\varepsilon}\overline{\partial}\right)\varphi(z,\overline{z}).\tag{174}$$

Focus on the holomorphic part and write the singular expansion

$$R[T(z)\psi(w)] \sim \sum_n (z-w)^n A_n(w). \tag{175}$$

Inserting it into the contour integral gives

$$\frac{1}{2\pi i}\oint_{C_w} \mathrm{d}z\varepsilon(z)R[T(z)\psi(w)] = \frac{\varepsilon''(w)A_{-3}(w)}{2} + \varepsilon'(w)A_{-2}(w) + \varepsilon(w)A_{-1}(w) + ... \tag{176}$$

Matching the primary transformation law, we find

$$A_{-3} = 0, \quad A_{-2} = h\psi, \quad A_{-1} = \partial\psi. \tag{177}$$

Therefore the singular OPE is

$$T(z)\psi(w) \sim \frac{h\psi(w)}{(z-w)^2} + \frac{\partial\psi(w)}{z-w}. \tag{178}$$

More generally, an OPE can include local operators and their descendants. If O_k is a primary appearing in O_1O_2, its leading term has powers

$$O_1(z,\overline{z})O_2(w,\overline{w}) \sim \sum_k C_{12}^k (z-w)^{h_k-h_1-h_2}(\overline{z}-\overline{w})^{\overline{h}_k-\overline{h}_1-\overline{h}_2}O_k(w,\overline{w}) + ... \tag{179}$$

4.5 4.5 Example: free scalar

Consider the free scalar action

$$S = \frac{g}{2}\int \mathrm{d}^2x\partial_\mu\varphi(x)\partial^\mu\varphi(x). \tag{180}$$

Its propagator in $d = 2$ is logarithmic:

$$\langle\varphi(x)\varphi(y)\rangle = -\frac{1}{4\pi g}\ln(x-y)^2 + \text{const.} \tag{181}$$

The equation of motion is

$$\partial^\mu\partial_\mu\varphi = 0 \quad \Rightarrow \quad 4\partial\overline{\partial}\varphi = 0. \tag{182}$$

Locally, one can write $\varphi(z,\overline{z}) = \varphi(z) + \overline{\varphi}(\overline{z})$. The holomorphic correlator is

$$\langle\varphi(z)\varphi(w)\rangle = -\frac{1}{4\pi g}\ln(z-w). \tag{183}$$

The logarithm means that φ itself is not an ordinary scaling primary, whereas

$$\langle\partial\varphi(z)\partial\varphi(w)\rangle = \partial_z\partial_w\left(-\frac{1}{4\pi g}\ln(z-w)\right) = -\frac{1}{4\pi g(z-w)^2}. \tag{184}$$

Thus $\partial\varphi$ is a candidate primary with weights $\left(h,\overline{h}\right) = (1,0)$. Check its OPE with $T(z)$. The canonical stress tensor is

$$T_{\mu\nu} = g\left(\partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}\eta_{\mu\nu}\partial_\rho\varphi\partial^\rho\varphi\right). \tag{185}$$

Since $\eta_{zz} = 0$,

$$T(z) = -2\pi T_{zz} = -2\pi g : \partial\varphi(z)\partial\varphi(z) : . \tag{186}$$

Normal ordering defines the coincident-point product in the path integral formalism (the following equality should be understood within the correlator):

$$T(z) = -2\pi g \lim_{w\to z}\left(\partial\varphi(z)\partial\varphi(w) + \frac{1}{4\pi g(z-w)^2}\right). \tag{187}$$

and in canonical formalism, an equivalent definition is putting the annihilation operators to the right of the creation operators.

Using Wick contractions and re-expanding around w gives

$$T(z)\partial\varphi(w) \sim \frac{\partial\varphi(w)}{(z-w)^2} + \frac{\partial^2\varphi(w)}{z-w}. \tag{188}$$

The double-pole coefficient is one, confirming $h = 1$.

4.6 4.6 Conformal Ward identity

Let C enclose all insertions. Deform it into small contours around the fields:

$$\begin{aligned}&\left\langle \frac{1}{2\pi i}\oint_C \mathrm{d}z\varepsilon(z)T(z)\varphi_1(w_1,\overline{w}_1)...\varphi_n(w_n,\overline{w}_n)\right\rangle \\ &= \sum_{j=1}^{n}\left\langle \varphi_1...\frac{1}{2\pi i}\oint_{C_{w_j}} \mathrm{d}z\varepsilon(z)T(z)\varphi_j\big(w_j,\overline{w}_j\big)...\varphi_n\right\rangle.\end{aligned} \tag{189}$$

The OPE at w_j yields

$$\Big(h_j\varepsilon'\big(w_j\big) + \varepsilon\big(w_j\big)\partial_{w_j}\Big)\langle\varphi_1(w_1,\overline{w}_1)...\varphi_n(w_n,\overline{w}_n)\rangle. \tag{190}$$

With the variation convention above, this contribution equals $-\delta_\varepsilon\varphi_j$ inside the correlator. Since the contour relation holds for suitable $\varepsilon(z)$, the conformal Ward identity is

$$\left\langle T(z)\prod_{i=1}^{n}\varphi_i(w_i,\overline{w}_i)\right\rangle = \sum_{j=1}^{n}\left(\frac{h_j}{\big(z-w_j\big)^2} + \frac{1}{z-w_j}\partial_{w_j}\right)\left\langle\prod_{i=1}^{n}\varphi_i(w_i,\overline{w}_i)\right\rangle. \tag{191}$$

Regular OPE terms give zero on each small contour. The formula assumes the usual behaviour of $T(z)$ at infinity.

4.7 4.7 Transformation of $T(z)$

Is $T(z)$ a primary? No. It is a quasiprimary with weights $\big(h,\overline{h}\big) = (2,0)$. Its self-OPE is

$$T(z)T(w) \sim \frac{c}{2(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{192}$$

Under a finite holomorphic change of coordinates $z' = f(z)$,

$$T'(z') = \frac{\partial f}{\partial z}^{-2}\Big(T(z) - \frac{c}{12}\{f,z\}\Big), \tag{193}$$

where the Schwarzian derivative is

$$\{f,z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2. \tag{194}$$

The central charge c represents a quantum effect and depends on the CFT. For a free boson, $c = 1$.

Bibliography